SUBSCRIPTION CREEP RADAR

A Power BI project report on recurring spend, subscription health, payments and usage

PROJECT REPORT

By
Arpit Kumar
Ankita Rani

Technology stack

Power BI Desktop • Power Query • DAX • Power BI Service • CSV

Reporting currency: Indian Rupees (INR)
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Document purpose and status

This report explains the business rationale, scope, design, implementation approach, analytical features, current deliverables, limitations and proposed next steps for Subscription Creep Radar. It is intended to support project evaluation, portfolio presentation and technical discussion.

Data integrity note The underlying dataset is described as learning/portfolio data. Unless replaced with verified operational data, the results must not be represented as actual company or customer performance.

Implementation status The README describes the project as developed so far, but also asks that the current PBIX model, exact field names, visual fields and final dashboard contents be verified. This report therefore distinguishes documented implementation from recommended validation and future work.

Executive summary

Subscription Creep Radar is a Power BI analytics solution designed to make recurring subscription costs and related activity easier to understand. Multiple individually low-cost subscriptions can accumulate into a material monthly expense. At the same time, subscription status, automatic renewal, payment outcomes and usage records are often spread across different datasets. The project brings these perspectives together in one analytical model.

The solution uses four related CSV datasets—Customers, Subscriptions, Payments and Usage—prepared in Power Query and connected through a one-to-many relational model. DAX measures calculate key indicators such as active monthly-equivalent spend, active subscriptions, customer count, payment totals and average payment amount. A three-page report explores subscription overview, subscription health, and usage/payment insights. Selected visuals are also pinned to a Power BI Service dashboard for at-a-glance monitoring.

The project was selected because it is a practical, recognisable business problem that demonstrates a balanced range of Power BI skills without requiring an overly complex enterprise architecture. It moves beyond a basic expense chart by combining financial, operational and behavioural indicators.

At a glance

Dimension	Project definition
Project name	Subscription Creep Radar
Primary platform	Microsoft Power BI
Supporting tools	Power Query, DAX, Power BI Service
Data format	Four CSV datasets
Currency	INR
Core analysis	Recurring spend, subscription status, payment behaviour and usage trends
Report structure	Three report pages plus a Power BI Service dashboard
Data status	Learning / portfolio demonstration data

1. Background and business problem

Subscription-based services are convenient because they convert purchases into recurring payments. However, convenience can also make spending less visible. A user or organisation may accumulate subscriptions across entertainment, productivity, cloud services, software and other categories. Each charge can look small in isolation, while the combined recurring commitment becomes significant.

The problem is not limited to the amount spent. A useful review also needs to understand which subscriptions remain active, which have been cancelled or paused, whether auto-renewal is enabled, what payment records show, and whether recorded usage changes over time.

1.1 Problem statement

Problem statement Subscription information, payment records and usage activity need to be viewed together to understand the scale and composition of recurring subscription spending, monitor subscription status and auto-renew behaviour, and explore payment and usage trends.

1.2 Why this problem matters

- Cost visibility: a consolidated view helps reveal the combined monthly-equivalent cost of active subscriptions.
- Subscription hygiene: status and auto-renew indicators help identify subscriptions that may deserve review.
- Category insight: category and plan comparisons show where subscriptions are concentrated.
- Payment monitoring: monthly payment trends and payment-status distribution provide a view of transaction activity.
- Usage context: usage trends add another dimension to the financial picture and can support future value-for-money analysis.

1.3 Why this project statement was chosen

The statement was chosen to sit in the useful middle ground between a simple dashboard exercise and an oversized enterprise project. It is familiar enough that an evaluator can understand the business need immediately, but rich enough to demonstrate data preparation, relational modelling, DAX, visual storytelling and publishing.

- It has a clear business question: where is recurring subscription money going, and what is happening to those subscriptions?
- It requires multiple related datasets rather than a single flat spreadsheet, making data modelling meaningful.
- It supports several analytical perspectives—financial, operational and usage-based—without requiring predictive modelling to make the report useful.
- It provides natural opportunities to demonstrate Power Query transformations, DAX filter context, one-to-many relationships, report interactions and Power BI Service.
- It can be expanded incrementally. The first version is achievable, while future enhancements such as churn definitions and time intelligence provide a credible roadmap.

The project is therefore both demonstrable and extensible: it can be completed to a presentable standard within a limited project window, while still offering meaningful technical discussion during an interview or evaluation.

2. Objectives and scope

2.1 Project objectives

1. Summarise monthly-equivalent spend across active subscriptions.
2. Track the customer base and total subscription population.
3. Monitor active, cancelled and paused subscription statuses.
4. Measure the number of subscriptions with auto-renew enabled.
5. Compare subscription counts and spending across categories and plans.
6. Analyse payment amounts, payment statuses and monthly transaction trends.
7. Explore recorded usage trends over time.
8. Publish a compact dashboard of important indicators in Power BI Service.
9. Demonstrate an end-to-end BI workflow from raw CSV data to an interactive report.

2.2 In scope

- Four CSV data sources and their integration into a Power BI data model.
- Data-type checks, data-quality review and appropriate standardisation in Power Query.
- One-to-many relationships between customers, subscriptions, payments and usage.
- Core DAX measures and a three-page report.
- A Power BI Service dashboard built from selected report visuals.

2.3 Out of scope for the current version

- Live integration with subscription providers, payment gateways or bank accounts.
- Automated cancellation, renewal or payment actions.
- Predictive churn modelling or customer lifetime value estimation.
- A verified production deployment using real company/customer data.
- A definitive churn-rate calculation without a defined period, denominator and business definition.

3. Users and practical use cases

The report is designed as an analytical aid. It does not itself cancel subscriptions or make financial decisions; it organises information so that a user or analyst can investigate patterns and decide what to review.

Potential user	Use case	Relevant report area
Individual / household budget reviewer	Review recurring commitments and identify high-spend categories.	Subscription Overview
Finance or operations analyst	Monitor recurring cost, subscription counts and payment activity.	Overview; Usage and Payment Insights
Subscription administrator	Review active, cancelled and paused records and auto-renew settings.	Subscription Health
Business manager	Get a quick summary of key subscription indicators.	Power BI Service dashboard
Data / BI evaluator	Assess data modelling, transformation, DAX and report design.	Entire solution

3.1 Example analytical questions

- What is the monthly-equivalent cost of all active subscriptions?
- Which subscription categories account for the largest share of recurring spend?
- How many subscriptions are active, cancelled or paused?
- How many subscriptions have auto-renew enabled?
- Which plans have the highest subscription counts?
- How do recorded payment amounts vary by month?
- What proportion of payment records fall into each payment status?
- How does recorded usage change over time?

4. Data sources and data model

4.1 Source datasets

Dataset	Role in the model	Example fields
Customers.csv	Customer details and unique customer identifiers.	CustomerID
Subscriptions.csv	Subscription status, plan, category and monthly-equivalent cost.	SubscriptionID, CustomerID, Status, Category, PlanName, CurrentMonthlyEquivalentINR
Payments.csv	Payment dates, amounts and payment outcomes.	SubscriptionID, PaymentDate, NetAmountINR, AmountBeforeDiscountINR, PaymentStatus
Usage.csv	Usage records associated with subscriptions and months.	SubscriptionID, UsageMonth, UsageHours

The working dataset previously contained approximately 350 customers, 1,050 subscriptions, 9,059 payment records and 10,960 usage records. These are historical working counts from the project notes and should be rechecked after any data refresh or file replacement.

4.2 Proposed relational structure



Relationships:

- Customers[CustomerID] → Subscriptions[CustomerID]
- Subscriptions[SubscriptionID] → Payments[SubscriptionID]
- Subscriptions[SubscriptionID] → Usage[SubscriptionID]

These are intended one-to-many relationships. The key in the one-side table must be unique, related columns must use compatible data types, and filter direction should be configured deliberately. An unnecessary direct Customers-to-Payments relationship should be avoided if it creates ambiguous filter paths.

4.3 Why separate tables are used

The four-table structure reflects distinct business entities and event types. Customers describe who holds subscriptions; Subscriptions describe the recurring service arrangement; Payments record financial transactions;

Usage records describe activity. Keeping these concepts separate reduces duplication and allows the report to analyse subscription-level attributes alongside multiple payment and usage records.

Modelling principle The subscription table is the central link between customer-level subscription information and the many payment and usage records associated with each subscription. Correct relationships are essential: a visually polished report can still produce misleading totals if the model is wrong.

5. Data preparation and quality approach

Power Query is used to import and prepare the CSV files before analysis. The intention is to make the data types, keys and labels consistent and to identify questionable records before they influence report values.

10. Import each CSV through Home → Get data → Text/CSV.
11. Set appropriate data types for identifiers, dates, currency amounts, usage hours and text fields.
12. Inspect null values and errors. Replace missing values only when the replacement is justified by the meaning of the field.
13. Check duplicate customer and subscription keys, especially on the one side of relationships.
14. Standardise category and status labels where inconsistent spelling or formatting would split the same business category.
15. Review date ranges, negative amounts and implausible usage values.
16. Apply changes and confirm that the model refreshes successfully.
17. Record material cleaning decisions so that the analysis remains explainable.

5.1 Quality checks that matter

Check	Why it matters
Unique dimension keys	Prevents ambiguous or invalid one-to-many relationships.
Consistent data types	Allows relationships, sorting and calculations to work correctly.
Nulls and errors	Avoids silent omissions or misleading replacements.
Status/category standardisation	Prevents one logical value from appearing as multiple categories.
Date validity and chronology	Supports reliable monthly analysis and time comparisons.
Amount and usage plausibility	Helps identify outliers or source-data issues for review.

6. Measures and analytical logic

DAX measures convert the prepared model into business indicators. The measures below are documented in the project README. Exact table names, field names and status values should be checked against the current PBIX model before formulas are reused.

6.1 Core measures

Total Monthly Spend

```
CALCULATE(
    SUM(Subscriptions[CurrentMonthlyEquivalentINR]),
    Subscriptions[Status] = "Active"
)
```

Sums monthly-equivalent cost for active subscriptions. This estimates current recurring monthly spend; it is not historical payment revenue.

Active Subscriptions

```
CALCULATE(
    COUNTROWS(Subscriptions),
    Subscriptions[Status] = "Active"
)
```

Counts subscription rows whose status is Active.

Total Customers

```
DISTINCTCOUNT(Customers[CustomerID])
```

Counts distinct customers in the current filter context.

Total Revenue

```
SUM(Payments[NetAmountINR])
```

Sums net amounts on payment rows. It may include failed transactions if their amount field is populated.

Total Payments

```
COUNTROWS(Payments)
```

Counts payment records; it does not necessarily count successful payments only.

Average Payment Amount

```
AVERAGE(Payments[NetAmountINR])
```

Calculates the average net amount per payment row; it may include failed transactions.

6.2 Optional successful-payment measure

```
Successful Payment Revenue =
CALCULATE(
    SUM(Payments[NetAmountINR]),
    Payments[PaymentStatus] = "Paid"
)
```

This measure should be used only if Paid is the exact successful status value in the source data. The business definition of revenue should also be agreed: the amount field and treatment of refunds, reversals or failed transactions can change the result.

6.3 Important metric distinctions

- Monthly spend is calculated from active subscriptions and their monthly-equivalent costs.
- Payment revenue is calculated from payment records and the selected amount field.
- Payment count may include failed payment records.
- A cancelled-subscription count is not, by itself, a churn rate.
- Abbreviated values such as 346.36K are display formats; they should not be mistaken for a different underlying unit.

Interpretation safeguard Do not label monthly-equivalent subscription spend as revenue. Do not label a cancelled count as churn rate. Clear metric definitions are part of the analytical design, not just wording.

7. Report design and features

7.1 Page 1 — Subscription Overview

Purpose: provide a high-level view of recurring subscription cost and the customer base.

- KPI cards: Total Monthly Spend, Active Subscriptions and Total Customers.
- Category analysis: monthly spend by subscription category.
- Category slicer, where configured, to support focused investigation.

This page answers: What is the monthly-equivalent spend on active subscriptions? How many active subscriptions and customers are represented? Which categories contribute most to recurring spending?

7.2 Page 2 — Subscription Health

Purpose: monitor the composition and status of the subscription portfolio.

- KPIs: Total Subscriptions, Active Subscriptions, Cancelled Subscriptions and Auto-Renew Subscriptions.
- Subscription status distribution.
- Subscription count by category.
- Subscription count by plan name.

This page answers: How many subscriptions are active, cancelled or paused? How many have auto-renew enabled? Which categories and plans contain the most subscriptions?

A cancelled count does not explain why a subscription was cancelled and does not establish a churn rate. A defensible churn rate requires a defined time period, denominator and business definition.

7.3 Page 3 — Usage and Payment Insights

Purpose: explore payment activity and recorded usage over time.

- KPIs: Total Revenue, Total Payments and Average Payment Amount.
- Revenue by month.
- Usage hours by month.
- Payment-status distribution.

This page answers: How do payment amounts vary by month? What is the average amount per payment record? How many payment records are present? What is the payment-status distribution? How does recorded usage change over time?

Before finalising the revenue visual, confirm whether it uses NetAmountINR or AmountBeforeDiscountINR and whether failed transactions are included. The visual title should describe the actual measure.

7.4 Interaction and usability

- Use report pages to move from summary to more focused analysis.
- Use slicers, where configured, to filter relevant categories or dimensions.
- Use visual tooltips and cross-filtering to inspect selected data points.
- Keep titles, units, labels and rounding clear.
- Use a bar chart when precise category comparisons matter; a donut chart is better suited to showing broad share-of-total composition.

8. Power BI Service dashboard

The dashboard is named Subscription Creep Radar Dashboard. It is created in Power BI Service by pinning selected report visuals. Its role is to provide a compact monitoring canvas, while the multi-page report remains the place for deeper exploration.

8.1 Documented KPI tiles

KPI tile	Previously recorded value
Total Monthly Spend	Approximately ₹346.36K
Active Subscriptions	817
Total Customers	350
Cancelled Subscriptions	164

These values reflect the dataset and filter context at the time of the last recorded refresh. They are not guaranteed current values and must be verified after refreshing the data.

8.2 Supporting visuals

- Monthly Spend by Subscription Category.
- Category share-of-spend donut chart (considered or included).
- Subscription Status.
- Revenue by Month.
- Payment Status Distribution.

The final dashboard should retain visuals that add distinct insight rather than repeating the same information. Dashboard tiles may not behave exactly like their original report visuals; use Go to report, where available, for detailed exploration.

8.3 Report versus dashboard

Power BI report	Power BI dashboard
Can contain multiple pages.	Uses a single dashboard canvas.
Supports detailed analysis and report interactions.	Summarises selected tiles at a glance.
Built primarily in Power BI Desktop.	Arranged in Power BI Service.
Best for exploration.	Best for monitoring key metrics.

9. End-to-end implementation approach

18. Define the business problem and the questions the report must answer.
19. Identify the required entities and prepare the four CSV source tables.
20. Import the data into Power BI and inspect the initial structure.
21. Use Power Query to set types, review quality issues and standardise values where appropriate.
22. Create the customer-to-subscription and subscription-to-payment/usage relationships.
23. Create and validate DAX measures against the intended business definitions.
24. Build the three report pages around distinct analytical questions.
25. Publish the report to the intended Power BI workspace.
26. Pin selected visuals to the Subscription Creep Radar Dashboard.

27. Reopen the report and dashboard, validate interactions and layout, and document known limitations.

This sequence follows a standard BI lifecycle: understand the question, prepare data, model it, calculate metrics, communicate findings and validate the published experience.

10. Current deliverables and validation status

Deliverable	Documented status / action
Four CSV source tables	Defined in the project; verify the current files and refresh counts.
Power Query preparation	Preparation workflow documented; record actual cleaning decisions.
Relational data model	Target relationships documented; verify uniqueness, data types and filter behaviour in PBIX.
DAX measures	Core measures documented; verify formulas and metric definitions in the current model.
Three-page report	Page structure and intended visuals documented; verify final visual fields and interactions.
Power BI Service dashboard	Dashboard name and previously recorded tiles documented; verify current tile set and layout.
PBIX and repository package	Save and organise final files; the README provides a suggested GitHub structure.

10.1 Final validation checklist

- ☐ All four CSV files load without errors.
- ☐ Data types are appropriate and relationship keys match.
- ☐ Keys on the one side of each relationship are unique.
- ☐ Relationships filter the intended tables without ambiguous paths.
- ☐ Nulls, duplicates and unusual values have been reviewed.
- ☐ Total Monthly Spend includes active subscriptions only.
- ☐ Total Customers counts distinct customer IDs.
- ☐ Subscription counts reconcile with source data.
- ☐ Revenue uses the intended amount field and handles failed payments as defined.
- ☐ Month labels sort chronologically; use Year-Month or a date table if multiple years are present.
- ☐ Slicers, tooltips and cross-filtering behave as expected.
- ☐ The PBIX is saved and published to the intended workspace.
- ☐ Dashboard tiles point to the intended visuals and layout persists after reopening.
- ☐ Sharing permissions are correct and no restricted information is exposed.

11. Limitations and risks

Limitation	Effect on interpretation	Mitigation / response
Learning / portfolio dataset	Results do not establish real-world business performance.	Label data clearly; replace only with verified and permitted data.
Monthly spend differs from payment revenue	The two metrics cannot be compared as if they were the same measure.	Use explicit definitions and accurate visual titles.
Payment sums may include failed transactions	A raw sum may overstate successful collections.	Create a successful-payment measure after confirming status values.
Cancelled count is not churn	A count alone does not show cancellation	Define a churn period, denominator and

	rate or timing.	business rule.
Values depend on refresh and filter context	Previously recorded KPI values may change.	Display refresh context where useful and revalidate after refresh.
Multiple years can create ambiguous month labels	Months may combine across years or sort incorrectly.	Add a date table or Year-Month field.
Dashboard tiles have limited interaction	A tile may not provide the same experience as the report visual.	Link users back to the report for detailed analysis.

12. Future enhancements and roadmap

The following enhancements are proposed rather than claimed as completed. They are ordered to improve correctness first, then analytical depth and user experience.

Priority	Enhancement	Expected benefit
1 — Foundation	Add a dedicated calendar/date table.	Reliable month, quarter, year and time-intelligence analysis.
1 — Foundation	Create successful-payment revenue excluding failed transactions.	A clearer measure of successful collections.
2 — Analysis	Add month-over-month and year-over-year comparisons where data supports them.	Makes changes and trends easier to interpret.
2 — Analysis	Add category share-of-spend percentages.	Shows each category's contribution to total recurring spend.
2 — Analysis	Define churn with a clear time window and denominator.	Turns status counts into a defensible retention metric.
2 — Analysis	Compare usage with subscription cost where meaningful.	Supports investigation of potential value-for-money patterns.
3 — Experience	Add report tooltips, drill-through pages and accessible labels.	Improves navigation, context and accessibility.
3 — Experience	Test report and dashboard layouts at different screen sizes.	Improves usability across devices and display sizes.
3 — Governance	Add a data dictionary for every field.	Makes the model easier to maintain, review and hand over.

12.1 Further ideas beyond the documented roadmap

If the project is extended beyond its current learning scope, possible next steps include a subscription renewal calendar, an exception page for subscriptions with auto-renew enabled, a detailed payment reconciliation view, and a configurable threshold to flag unusually expensive subscriptions. These ideas would require suitable fields and clear business rules in the source data; they are not part of the documented current implementation.

13. Suggested repository structure

```
subscription-creep-radar/
├── README.md
├── report/
│   └── Subscription Creep Radar.pbix
├── data/
│   ├── Customers.csv
│   ├── Subscriptions.csv
```

```
│   ├── Payments.csv
│   └── Usage.csv
└── screenshots/
    ├── subscription-overview.png
    ├── subscription-health.png
    ├── usage-payment-insights.png
    └── dashboard.png
```

This is a suggested structure only; it does not confirm that the files have already been placed in these folders. Before publishing a public repository, confirm that the PBIX and source data contain no personal, confidential or company-restricted information. Clearly label synthetic data as synthetic.

14. Expected learning outcomes and skills demonstrated

- Data ingestion: importing multiple CSV sources into Power BI.
- Data transformation: applying Power Query data-type, null, duplicate and standardisation checks.
- Data modelling: designing one-to-many relationships and avoiding ambiguous filter paths.
- DAX: using CALCULATE, SUM, COUNTROWS, DISTINCTCOUNT and AVERAGE in measures.
- Analytical reasoning: distinguishing recurring monthly cost from historical payment activity.
- Visual communication: designing pages around business questions and selecting appropriate chart types.
- Publishing: creating a Power BI Service dashboard from report visuals.
- Validation: checking calculations, relationships, refreshes, interactions and published layout.

15. Interview-ready project explanation

60-second explanation "Subscription Creep Radar is a Power BI project that analyses recurring subscription spending, subscription status, customer counts, and payment and usage trends. I used Power Query to prepare four CSV datasets, created relationships between customers, subscriptions, payments and usage, and built DAX measures for indicators such as total monthly spend, active subscriptions, total customers, revenue and average payment amount. I designed a three-page report covering Subscription Overview, Subscription Health, and Usage and Payment Insights, then published selected visuals to Power BI Service as a dashboard. The key design consideration was to distinguish monthly-equivalent recurring spend from payment revenue so that the metrics are interpreted correctly."

15.1 Questions to prepare for

- Why did you separate Customers, Subscriptions, Payments and Usage into different tables?
- What makes each relationship one-to-many, and how did you validate the keys?
- What is the difference between a measure and a calculated column?
- How do CALCULATE, SUM, COUNTROWS, DISTINCTCOUNT and AVERAGE work in your measures?
- Why is monthly-equivalent subscription spend different from payment revenue?
- How would you exclude failed payments from successful revenue?
- Why is cancelled-subscription count not the same as churn rate?
- How did you handle missing values, data types and duplicate keys in Power Query?
- What is the difference between a Power BI report and a Power BI dashboard?
- What would you improve first if the project were used with real operational data?

16. Conclusion

Subscription Creep Radar demonstrates how Power BI can turn several related datasets into a coherent view of recurring subscription costs, subscription health, payment activity and usage trends. Its value lies in combining these perspectives in one model and organising them around practical questions that a user or analyst can investigate.

The current scope is deliberately focused: a four-table model, a set of core DAX measures, three analytical report pages and a Power BI Service dashboard. The next stage should prioritise validation of the current PBIX, precise revenue definitions and reliable date modelling before adding more advanced analysis. With those foundations in place, the project has a clear path from a portfolio demonstration to a more mature subscription-monitoring solution.

— End of Project Report —